

degree in $\geq 50\%$ of FU and as occasional if reported in $< 50\%$. Crude incidence rates of vaginal functioning problems during sexual activity (dryness, shortness, tightness and pain) at baseline and during FU were calculated if reported as “quite a bit” or “very much” (substantial symptoms). Association between vaginal shortening and tightening was evaluated with Spearman's rank correlation, pooling observations over all FU ($N=2669$).

Results: After the diagnosis of cancer, but prior to treatment, 230/1027 (22.4%) of patients reported some degree of sexual activity. During FU (median 36 months), 433/1044 (41.5%) patients reported no sexual activity, 128 (12.3%) occasional and 483 (46.3%) frequent sexual activity. Prior to treatment, vaginal dryness, shortening, tightening and pain during intercourse were reported by 7.1%, 2.9%, 4.8% and 10.5% of patients, respectively. During follow-up these sexual functioning problems (crude incidence) increased to 38.4%, 36.4%, 34.2% and 33.5%, respectively. The median sexual enjoyment over all FU for individual patients was “quite a bit” / “very much” in 55.6%. Vaginal shortening and tightening during intercourse were significantly correlated (Spearman's rho 0.661; 19% tightening alone, 25% shortening alone, 56% both).

Conclusion: More than half of the patients responding to the EORTC questionnaire are sexually active after treatment and the majority report frequent sexual activity. Sexual functioning problems are reported in around one third of patients and sexual enjoyment is compromised in almost half of sexually active patients. This underlines the importance of efforts directed towards prevention of vaginal morbidity and sexual rehabilitation after treatment.

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Comparison of Concurrent Cisplatin Chemoradiotherapy Plus Adjuvant Chemotherapy Versus Concurrent Chemoradiotherapy Alone in Locally Advanced Cervical Cancer



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Purpose/Objective(s): According to NCCN guidelines, concurrent chemoradiation (CCRT) is the standard care for locally advanced cervical cancer (LACC) patients, but the benefit of adjuvant chemotherapy (ACT) after CCRT for LACC patients is still controversial. This study is designed

to answer the question by comparing the efficacy and toxicities of concurrent chemoradiation (CCRT) with adjuvant chemotherapy (ACT) (CCRT+ACT) versus concurrent chemoradiation (CCRT) alone in LACC.

Materials/Methods: Patients with stage IB2,IIA2,IIIB-IVA (2009 FIGO staging) cervical cancer were recruited from June 2013 to June 2018 at 4 cancer centers in China. The patients were randomly assigned to two arms: CCRT +ACT and CCRT alone. All patients were treated with IMRT (external-beam radiotherapy up to 50.4 Gy) with concurrent cisplatin, followed by intra-cavity brachytherapy. The chemotherapy consisted of concurrent cisplatin (40 mg/m² weekly) during IMRT, and 3 cycles of adjuvant CT with TP (docetaxel 75 mg/m², day 1, and cisplatin 25 mg/m², day 1-3 or carboplatin AUC 4-5, day 1) every 3 weeks. The primary endpoint was progression free survival (PFS), and the secondary endpoints included overall survival (OS), locoregional recurrence rate, distant metastasis rate, and toxicities. Survivals were calculated with Kaplan-Meier method. The differences between survivals were calculated with Log-rank test. Toxic effects were analyzed using the χ^2 test. All statistical tests were two-sided.

Results: A total of 176 patients were recruited in the study and 161 were eligible for intent-to-treat analysis. 82 patients were assigned to the CCRT+ACT arm and 79 in the CCRT alone arm. With a median follow-up of 60.1 months, the 5-year PFS and OS of all the patients were 68.32% and 72.05%, respectively. The 5-year PFS was 75.61% in the CCRT+ACT arm, and was 60.76% in the CCRT arm. There was significant difference in PFS between the two arms ($P=0.043$). The 5-year OS was 76.83% in the CCRT+ACT arm and was 67.09% in the CCRT arm. There was no significant difference in OS between the two arms ($P=0.169$). There was significant difference between the CCRT+ACT and CCRT arms in distant alone recurrence ($p=0.048$), but not in locoregional recurrence rate ($p=0.752$). There were significantly higher acute systemic side effects in the CCRT+ACT arm, especially the incidence of grade 3-4 hematologic and gastrointestinal reactions ($P=0.000$). Most of locoregional recurrence and distant metastases occurred with staging IIIB-IVA patients in the CCRT alone arm.

Conclusion: Adjuvant chemotherapy after concurrent cisplatin chemoradiation for LACC patients improves PFS but not improves OS. It reduces distant metastasis but is correlated with poorer response to locoregional recurrence. There were more significant toxicities during adjuvant chemotherapy but it was well tolerated by patients. Further research and longer follow up is needed, especially for the staging IIIB-IVA LACC patients.

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Radiotherapy vs. Trans-Oral Robotic Surgery for Oropharyngeal Squamous Cell Carcinoma (OPSCC): Results of a Randomized Trial



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Purpose/Objective(s): The incidence of oropharyngeal squamous cell carcinoma (OPSCC) has risen rapidly, due to an epidemic of human papillomavirus (HPV) infection. Radiation therapy (RT) has historically been the standard treatment, but transoral robotic surgery (TORS) has surpassed RT in the US as the most common approach, based on assumptions of reduced toxicity or improved quality of life (QOL). No randomized trials have previously compared these treatments.

Materials/Methods: The ORATOR trial (NCT01590355) enrolled patients with T1-T2 N0-2 (≤ 4 cm) OPSCC amenable to TORS. We randomly assigned patients, stratified by p16 status, to RT (70 Gy/35 fractions, or as chemoradiotherapy [CRT] if N1-2) vs. TORS ($\pm$ adjuvant RT or CRT based on pathology). The primary endpoint was a definitive comparison of swallowing QOL at 1-year using the MD Anderson Dysphagia Inventory (MDADI), powered to detect a 10-point improvement (a clinically-meaningful change [CMC]) in the TORS arm. Secondary endpoints included adverse events (AEs), other QOL outcomes [including EORTC Head and Neck-35 (H&N-35) scale and other scales], overall and progression-free survival (OS, PFS). Herein we report primary and secondary outcomes with a post-hoc analysis of outcomes by treatment intensity.

Results: Between 2012 and 2017, 68 patients were randomized ($n=34$ in each arm), at 6 centres in Canada and Australia. Median age was 59 years; 87% were male. Primary tumors sites were palatine tonsil (74%) or base of tongue (26%). Arms were well-balanced for baseline factors, including p16 status (88% in each arm). Median follow-up was 27 months. MDADI scores at 1-year were statistically superior in the RT arm (mean $\pm$ SD: 86.9 ± 11.4 vs. 80.1 ± 13.0 in the TORS arm; $p=0.042$), but not meeting the definition of a CMC. For other QOL metrics, outcomes were similar at 1-year. Percutaneous feeding tube rates at 1-year were 3% ($n=1$) vs. 0% respectively. Rates of treatment-related grade ≥ 2 AEs were similar (91% vs. 100%, $p=0.24$), with more neutropenia, hearing loss, constipation and tinnitus in the RT arm and more trismus in the TORS arm (all $p<0.05$). There was one TORS bleeding-related death. The arms were similar in OS ($p=0.88$) and PFS ($p=0.63$). Longitudinal QOL analysis over time confirmed the statistical superiority of RT in the MDADI scores ($p<0.001$), with average differences below the threshold of a CMC. In the RT Arm, 1-year MDADI scores based on treatment intensity were as follows: RT alone 89.5 ± 6.2 ; CRT alone 88.0 ± 11.5 RT; RT/CRT with salvage surgery 68.0 ± 7 (ANOVA $p=0.044$). In the TORS arm, the 1-year MDADI scores by treatment intensity were: TORS alone 82.8 ± 10.5 ; TORS with RT 78.5 ± 11.0 ; TORS with CRT 80.4 ± 19.6 (ANOVA $p=0.760$).

Conclusion: RT had superior swallowing QOL scores at 1-year compared to TORS, but the difference was not a CMC. Toxicity patterns differed between the arms. Patients with OPSCC should be counselled on the advantages and disadvantages of both modalities.

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A Phase II Trial of Alternative Volumes of Oropharyngeal Irradiation for Deintensification (AVOID): Omission of the Resected Primary Tumor Bed for Squamous Cell Carcinoma of the Oropharynx



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Purpose/Objective(s): Treatment of human papillomavirus (HPV)-associated oropharyngeal squamous cell carcinoma (OPSCC) with initial surgical resection followed by adjuvant radiotherapy yields high cure rates, but can cause long-term toxicity. This phase II trial evaluated whether omitting high dose radiation to the resected primary tumor bed while targeting only the regional neck nodes, could maintain high local control rates while reducing late sequelae.

Materials/Methods: Eligible patients were those appropriate for adjuvant RT after surgical resection for p16-positive, pathologic stage III-IVa (7th edition) p16-positive OPSCC. Pathologic inclusion criteria included: T1-T2 disease, any N1-2, negative resection margins, and no perineural or lymphovascular invasion. Patients received 60-66 Gy RT over 30-33 fractions to the regional nodes of the bilateral neck, with active omission of high-dose RT to the resected primary tumor bed. Concurrent chemotherapy was given for patients with extranodal extension. The primary end point was 2-year locoregional control.

Results: Sixty patients were enrolled. Thirty-two (53%) received intensity-modulated radiation therapy (IMRT), 27 (45%) received proton therapy, and 1 (2%) received a 50:50 hybrid IMRT:proton plan. Average follow-up was 2 years, with local control of 98%, regional control of 100%, distant control of 97%, and overall survival of 100%. Median dose delivered to the avoided primary site was 37 Gy (40 Gy for IMRT, 34 Gy for proton RT, $p=0.03$). There was only 1 local recurrence, occurring 9 months after completion of RT, which was successfully salvaged via surgical resection. Two (3%) patients experienced post-RT primary site tissue necrosis, both of whom experienced complete resolution with conservative management. Percutaneous endoscopic gastrostomy (PEG) tube dependence rates were: 0 patients (0%) during RT, and 1 patient (2%) at time of last follow-up.

Conclusion: Avoidance of radiation to the resected primary tumor bed appears safe in well-selected patients with HPV-associated OPSCC. This approach resulted in a low rate of treatment-associated complications, and is worthy of further study as a strategy for deintensification.

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Recurrence and Quality-of-Life Following Elective Nodal Volume and Dose De-Escalation for Oropharyngeal and Laryngeal Cancer: Initial Results from the Infield Trial



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Purpose/Objective(s): The required elective nodal dose and volumes for head and neck intensity-modulated radiotherapy (IMRT) have largely been extrapolated from conventional radiotherapy fields. The combination of CT and PET-CT imaging with surgically established patterns-of-nodal-spread provides an opportunity to meaningfully reduce the elective dose and volume, with a significant gain in acute and long-term morbidity. In